

Utkarsh Potdukhe

AI Engineer | LLM Applications | AI Automation | Full Stack Development

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Portfolio: utkarshpotdukhe.github.io/utkarsh-portfolio-v3

Summary

AI Engineer with 2+ years of experience building **LLM applications**, **AI agents** and **workflow automation**. Delivered **70+ automation projects** using **n8n** and **Python**, and built **RAG systems**, **agentic workflows** and **LLM gateways** with a strong focus on retrieval quality, access control, evaluation, observability and cost. **Led and mentored 2 interns**. Comfortable across the stack with **Next.js**, **React.js** and **Node.js**. Looking for AI Engineer, LLM Engineer or AI Automation roles where I can take AI systems from prototype to production.

Technical Skills

AI and LLM: LLM Applications, RAG, Agentic RAG, AI Agents, Multi Agent Systems, LangChain, LangGraph, OpenAI API, Prompt Engineering, Model Routing, LLM Evaluation, RAGAS

Retrieval: Hybrid Search, BM25, Vector Databases, Embeddings, Reranking, Semantic Caching

Automation: n8n, Zapier, Workflow Automation, Voice AI Agents, ElevenLabs, Zoho CRM, Apollo

Languages: Python, JavaScript, TypeScript, SQL, HTML5, CSS3

Frameworks: FastAPI, Node.js, React.js, Next.js, Redux, GraphQL

Infrastructure: Docker, REST APIs, Webhooks, JWT, RBAC, Redis, PostgreSQL, OpenTelemetry, Jaeger, Prometheus, Langfuse, Git, GitHub, CI/CD

Experience

Ecommerce Collections

Apr 2026 to Present

AI Automation Developer

Remote (USA)

- Designed **Retrieval Augmented Generation (RAG)** solutions on **vector databases** and structured knowledge bases so teams get grounded answers from company data
- Built a **lead intelligence platform** that researches prospects, enriches and qualifies them, and syncs the results to the CRM
- Developed **competitor intelligence automation** for continuous market monitoring and strategic business reporting
- Architected **multi agent automation systems** connecting **LLMs**, APIs, databases and business applications
- Built an **AI product listing platform** in **n8n** that generates SEO optimized titles, descriptions, bullet points, metadata, keywords and marketplace ready images using LLMs and **AI image generation** models
- Developed a **Google Search Console automation** that tracks keyword rankings, position changes, clicks, impressions, CTR and indexing status, and sends automated SEO insights and alerts

Progress with AI

Jul 2024 to Apr 2026

AI Automation and Web Developer

Pune, India

- Built and deployed **AI agents** for repetitive business tasks, **reducing manual workload by 40%**
- Delivered **full stack automation** projects, including self running **voice systems** that replaced manual work
- Integrated AI agents with internal systems through **n8n** and **API workflows** for automated decision making
- Developed internal web tools and real time monitoring dashboards with **Next.js** and **React.js**
- Improved React.js application performance by about 30%** using **Redux** and **GraphQL**
- Led **code reviews**, maintained shared UI libraries and contributed to **CI/CD** testing pipelines

Lovenspire

Nov 2025 to Apr 2026

AI Automation Engineer (Part Time)

Remote (USA)

- Automated **inventory documentation** by pulling product and stock data from **Amazon backend APIs**, **scraping** storefront data and syncing **Shopify** store data into one up to date inventory report
- Built an **AI SEO optimization** workflow that combines **Helium 10** keyword research with **LLMs** to write titles, bullet points, descriptions and backend search terms that follow brand rules and Amazon listing requirements
- Automated **AI image generation** for product listings, producing listing images that match brand guidelines and marketplace image requirements
- Improved **organic search ranking** of products through keyword rank tracking and ongoing **SEO optimization**, while keeping every listing within Amazon seller policies to protect account health

10xgrow.ai

Nov 2025 to Dec 2025

AI Agent Developer (Part Time)

Remote (USA)

- Developed production **AI voice agents** with **n8n** and a **Python** backend, covering real time speech processing, conversational AI, workflow orchestration, API integrations and CRM sync

Projects

Enterprise Knowledge Copilot | *Python, FastAPI, LangChain, Vector DB, BM25, Redis, JWT, RAGAS, OpenTelemetry*

- Built and deployed a role aware **RAG copilot** over company policy, handbook, technical and compliance documents, using **hybrid retrieval** (BM25 plus vector search) with a **reranker** on top
- Tagged every chunk with role metadata (employee, manager, admin) and applied **RBAC filtering** from **JWT** claims before any context reaches the LLM, so restricted content never enters the prompt
- Added a **semantic cache** keyed by user role, which cuts repeat query cost without serving a manager's answer to an employee
- Every answer cites its source chunks; evaluated **Recall@5**, **MRR**, **faithfulness** and unauthorized retrieval rate on a hand written golden Q&A set with **RAGAS**, and tracked **p95 latency** and cost per answer with **Jaeger** traces and **Prometheus**

Agentic Customer Operations Platform | *Python, LangGraph, FastAPI, PostgreSQL, OpenAI API*

- Designed a stateful support agent as a **LangGraph** state machine with tools for order lookup, policy lookup, ticket creation, refund drafting and human escalation
- The LLM only proposes actions; **deterministic policy gates** in code (refund window, amount limits) decide what runs, and refunds, cancellations and account changes wait for **human approval**
- Made ticket and refund actions **idempotent** so a retried request never creates a duplicate, and stored **durable checkpoints** so interrupted workflows resume safely
- Built **safety evals** with normal and adversarial requests, measuring tool selection accuracy, unauthorized action rate, escalation precision, task completion and cost per resolved case, with **tool level tracing**

LLM Gateway with Intelligent Model Routing | *Python, FastAPI, LiteLLM, Redis, Langfuse, Docker*

- Developed a **multi provider LLM gateway** behind one endpoint that routes each request to a small, medium or premium model by task complexity, **latency SLO** and tenant budget
- Implemented auth, per tenant **rate limits**, budget checks, response caching, timeouts with **retry and backoff**, and automatic **fallback** to a second provider
- Logged model, tokens, latency and cost for every request and built a **cost dashboard** per tenant and model tier with p50 and p95 latency SLOs
- Compared routed traffic against an all premium baseline on a fixed benchmark with a quality bar set before the test, so every **cost saving** is measured against task success, not just price

Agentic RAG Research System | *Python, LangGraph, BM25, Vector DB, Reranker, RAGAS*

- Implemented a **multi hop research copilot** where a **query planner** breaks complex questions into subqueries, retrieves and reranks evidence for each hop, and checks it before moving on
- Added a **faithfulness check** before every answer so the system **abstains** instead of guessing when the evidence does not support a claim
- Wrote a labeled eval set with gold sources before any code and measured **Hit@K**, **Recall@K**, **MRR**, **nDCG**, faithfulness, citation support, p95 latency and cost per query, with traces for each plan, hop and rerank step

AI Lead Qualification and Routing System | *n8n, OpenAI, Zoho CRM, Google Sheets, Apollo*

- Automated the full lead lifecycle from capture to **AI qualification**, product matching and salesperson assignment in real time, replacing manual lead handling
- Designed an AI qualification engine with **structured JSON output** that detects intent, prioritizes high value prospects and routes relevant and irrelevant leads through separate paths
- Matched each lead to the best fit salesperson by product expertise and availability, with internal alerts and HTML email notifications
- Synced enriched lead data to **Zoho CRM** and Google Sheets, **cutting lead response time by about 90%** with zero manual qualification

Voice Based Appointment Booking System | *n8n, ElevenLabs, Cal.com*

- Built a **voice scheduling agent** that lets callers book appointments over the phone with no human support, checking live availability and confirming bookings automatically

Google Maps Lead Generation (No Paid APIs) | *n8n, JavaScript, Web Scraping, Google Sheets*

- Built a **lead generation workflow** that scrapes business details from Google Maps results and extracts emails, phone numbers and business data with **regex parsing** and custom **JavaScript** nodes

Automated Proposal and Sales Assistant Workflows | *n8n, GPT 4o, Google Slides, Gmail, SMS*

- Created a **proposal automation** system that turns form inputs into ready to share Google Slides proposals sent through Gmail, and a **24/7 sales assistant** that captures leads and sends follow up emails and SMS automatically

Education

Government College of Engineering, Amravati

Bachelor of Technology in Electrical Engineering; CGPA: 7.42

2024

Amravati, India